

# SANJUKTA APARNA VENKATACHALAM

Embedded Software & Firmware Engineer · C · ARM Cortex-M · FreeRTOS · Bootloaders · FPGA/RTL

United Kingdom | 07810 891517 | [sanjuktaa23@gmail.com](mailto:sanjuktaa23@gmail.com) | [LinkedIn](#) | [Github](#) | [Portfolio](#)

Right to work: UK Student visa, moving to the Graduate Route (2 years) – no sponsorship required

## PERSONAL STATEMENT

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Embedded software engineer with an MSc in Embedded Systems Engineering (Distinction, 74%) from the University of Leeds. Built bare-metal and FreeRTOS firmware in C for ARM Cortex-M4F – register-level drivers, a custom UART bootloader with CRC-framed firmware update – and Verilog RTL on FPGA, and validate everything on live hardware with logic analysers, SWD and SEGGER SystemView. Top-20-of-800 finalist, NESIA National Innovation Challenge.

## SKILLS

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**Embedded & Architecture:** ARM Cortex-M4F (exception model, NVIC, MSP/PSP, memory map, MPU); bare-metal register-level programming, FreeRTOS, bootloaders and internal flash programming, DMA, interrupts, timers/PWM, RTC, low-power modes

**Protocols:** SPI, I2C, CAN, UART/USART (register + HAL level); LoRa 868 MHz, Wi-Fi

**Languages:** C, Embedded C, C++, Python, Verilog RTL, ARM Assembly, MATLAB

**FPGA/RTL:** Verilog RTL and testbenches, FSM design, Intel Cyclone V, Quartus Prime, ModelSim

**Debug & Tools:** Saleae logic analyser, oscilloscope, SWD/JTAG (ST-Link), SEGGER SystemView, ITM/SWV; GCC ARM toolchain, custom linker scripts, CMSIS, STM32CubeIDE, Keil µVision, Git/GitHub

**Hardware & Practice:** Hardware bring-up & schematic reading, sensor integration (IMU/MPU6050) with Kalman filtering, requirements-first validation

**Professional:** Teamwork, Technical research, Problem-solving, Communication, Time management, Adaptability

## KEY ENGINEERING PROJECTS

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### FlashForge - Custom STM32 Bootloader with UART Firmware Update

[GitHub](#)

*Embedded C · STM32 Cortex-M4F · linker scripts · internal flash · UART · Python host tool*

- Designed a custom bootloader across 10 milestones: custom linker scripts splitting flash between bootloader and application, VTOR relocation and a three-step jump (stack pointer, vector table, reset handler) into the application.
- Built a framed UART update protocol with CRC-32 (STM32 hardware CRC unit), image-header validation and a register-level flash driver; wrote a Python host tool (`--makeimg / --ping / --flash`) to build and send images
- Debugged a handover fault where interrupts left disabled across the jump hung `HAL_Delay`; fixed by clearing pending NVIC interrupts and re-enabling IRQs before the jump.
- Demonstrated failsafe recovery from a deliberately interrupted update.

### Broken Power-Line Detection over LoRa - MSc Dissertation

[GitHub](#)

*ESP32 · RFM95W 868 MHz · Raspberry Pi · Python / Flask · MATLAB*

- Designed and built a distributed LoRa fault-detection network end to end - ESP32 nodes relaying events over an 868 MHz link to a Raspberry Pi gateway for segment-level localisation, from MATLAB link-budget modelling to hardware.
- Implemented the wireless protocol from scratch - custom packet header, node addressing, private sync word, interrupt-driven receive with a heartbeat and miss-limit scheme for silence-based fault detection.
- Wrote the Python gateway service and Flask dashboard for live node status, event logging and email alerts; validated against requirements before build to avoid rework.

### Cortex-A9 NEON SIMD DSP Acceleration

[GitHub](#)

*ARM Cortex-A9 · NEON SIMD intrinsics · C · DE1-SoC*

- Bare-metal DSP suite (audio/SDR FIR, FFT, Kalman, image median filter) benchmarking scalar C against NEON vectorised kernels for correctness and speed.
- Vectorised the SDR FIR filter with NEON for a 5x speed-up over scalar, then profiled a median-filter case that ran at 0.75x, pinpointing the cause as branch divergence and non-contiguous memory access.

## Hardware-Accelerated Pre-Trade Risk Filter

[GitHub](#)

Verilog RTL · Intel Cyclone V · Quartus Prime · ModelSim

- FPGA filter that vets keypad-entered trade orders against 10 hardwired risk rules (quantity, price bounds, halted stock) and returns a PASS/FAIL verdict in ~20ns; verified in ModelSim before Cyclone V bring-up.
- Built the timing and display path - a hardware timer, a 5-state control FSM, binary-to-BCD conversion and a seven-segment driver presenting the verdict.

## ADDITIONAL FIRMWARE WORK

- **Register-level driver library** – GPIO, RCC, SPI, I2C, USART, EXTI and DMA drivers written from the reference manual with no HAL (blocking, interrupt and slave-mode APIs; DMA memory-to-peripheral and memory-to-memory with direct NVIC register access), each verified against live bus traffic on a logic analyser.
- **FreeRTOS on Cortex-M4** - integrated the kernel into a from-scratch STM32 project (GCC/ARM\_CM4F portable layer, heap scheme, linker settings); built multi-task applications with priority pre-emption and ISR-driven priority switching, traced live over UART with SEGGER SystemView.
- **Real-Time IMU Sensor Fusion** - sampled an MPU6050 over I2C on a data-ready interrupt, fused accelerometer/gyroscope data with a Kalman filter under a semaphore/ISR FreeRTOS architecture and drove PWM output from the estimate; tuned live in STM32CubeMonitor.
- **Advanced STM32 peripherals** - CAN in loopback and normal modes, timer-driven PWM and input capture, RTC alarms requiring the backup-domain unlock sequence, and WFI/WFE low-power modes.

## EDUCATION

### MSc Embedded Systems Engineering - University of Leeds, UK

Sep 2025 – Sep 2026

Distinction (74%)

**Relevant Modules:** Embedded Microprocessor System Design, FPGA Design for SoC, Control Systems Design, Wireless Communication System Design, Data Communications and Network Security, Medical Electronics and E-Health

### B.E. Electrical & Electronics Engineering - Alagappa Chettiar Govt. College, India

Nov 2021 – Apr 2025

CGPA 8.13 / 10

## WORK EXPERIENCE

### Consulting Project – DAP Technologies Co., Ltd

Aug 2026 – Sep 2026

Recommended an API-first integration layer for a client's aviation and eVTOL supply-chain platform. Shaped the system architecture and data-access model; presented the final pitch to the client.

### Engineer Tomorrow Scheme - Parker Hannifin, UK

Feb 2026 – Apr 2026

Selected onto a competitive industry scheme: designed a conceptual filtration system in CAD, applying engineering design and optimisation principles, and supported peers through technical problem-solving.

### Engineering Intern - Neyveli Lignite Corporation (NLC) India Ltd.

Jun 2024

Placed on a 210 MW+ thermal generation site: studied turbine control, substation automation and SCADA monitoring alongside senior engineers, analysing real-time fault-protection schemes in live operation.

### Embedded / IoT Intern - Verzeo

Oct 2022 – Nov 2022

Developed Arduino and ESP32 IoT prototypes with cloud integration for remote monitoring and data visualisation, validating reliable data transmission across each build.

**Part-time** (University of Leeds conferences, Catering, First Direct Arena, Primark) alongside MSc study.

## ACHIEVEMENTS & LEADERSHIP

- **Top 20 of 800 teams** at the NESIA National Engineering Innovation Challenge.
- **Outstanding Poster Presentation**, International Leadership Summit, Leeds.
- **1st and 2nd Prize** in consecutive years, College Project Expo (Thiran).
- **Event Head**, EEE Association - three public events, 500+ attendees. **Associate Student Member**, IEEE Women in Engineering. **Volunteer**, National Service Scheme.